INTERIM REPORT

WHO INFLUENZA CENTRE

LONDON

SEPTEMBER 2005

CHARACTERISTICS OF HUMAN INFLUENZA AH1N1, AH3N2 AND B VIRUSES ISOLATED FEBRUARY TO JULY 2005

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Of some 800 influenza viruses isolated in 34 countries since February 2005, approximately half (49%) were AH3N2, 17% were AH1N1 and 34% were influenza B (Table 1). Influenza B viruses accounted for 47% of the viruses isolated during March and April, of which 75% were B/Shanghai/361/2002-like (Yamagata lineage), although the relative prevalence of the two lineages varied in different countries. No H1N2 viruses were received for characterization since February 2004.

AH1N1 viruses

The majority of H1N1 viruses were antigenically closely related to the current vaccine strain A/New Caledonia/20/99 and similar to the more recent reference viruses A/Netherlands/128/2004 (MDCK) and A/Thessaloniki/24/2005 (egg) (Table 2). Most of the HA and NA sequences were similar to those of H1N1 viruses isolated during the previous year (red in Figures 1 and 2, Table 3).

AH3N2 viruses

Of some 100 H3N2 viruses tested during April, the majority (approximately 90%) gave patterns of HI reactivity similar to those of the egg-grown A/California/7/2004 or A/New York/55/2004 vaccine strains or the MDCK-grown A/Shantou/1219/2004 or A/Finland/486/2004 reference viruses (Table 4). Ferret antisera against these viruses did not discriminate between the isolates and less than 10% of the isolates gave low HI titres (relative to homologous titres) with all the antisera included in the tests. Subsequently, however, approximately 25% of some 250 viruses tested during June to August gave low HI titres with all the reference ferret antisera. These included, in particular, recent isolates from Hong Kong and South Africa, but also viruses recently received from a number of European countries (bottom of Table 4). Antisera to some more recent isolates, representative of distinguishable HA phylogenetic groups (see below), including MDCK-grown viruses A/Baden-Wurttemburg/38/2005 and A/Hong Kong/4355/2005 and the egg-grown viruses A/Brisbane/3/2005 (from Melbourne) and A/New Hampshire/14/2005 (from CDC, Atlanta), did not readily differentiate these viruses from earlier isolates, such as A/Finland/486/2004 or the prototype vaccine strain A/California/7/2004 (Table 5). As observed previously, HI tests using human (or guinea pig) red blood cells gave less marked differences in HI titres than turkey cells and it is apparent that differences in HI titres are at least in part due to differences in receptor binding.

The majority of HA sequences of 2005 isolates were close to that of A/California/7/2004 and characterized by the amino acid changes K145N, Y159F, S189N and S227P relative to the HA of A/Wyoming/3/2003. A number of variants were distinguished among more recent isolates and the HAs of many viruses isolated during April to July tended to fall within one of two phylogenetic groups (Figure 3):

1. One characterized by amino acid changes V112I and K173E. Other changes which differentiated variants included: N145S (and D188Y or A196T), represented by A/Baden Wurttemberg/38/2005 (A/BW/38/05); R201K (in e.g. A/Austria/209122/2005) and S193N and I214V (in e.g. A/Hong Kong/7728/2005) (blue in Figure 3, Table 6)
2. The other, characterized by S193F and D225N, was typical of many recent isolates from Hong Kong, such as A/Hong Kong/4355/2005 (red in Figure 3).

The NAs of these viruses were in general similar to those of A/California/7/2004 and A/New York/55/2004 (Figure 4, Table 6).

Influenza B viruses

The majority of B/Yamagata-lineage (HA) viruses were antigenically closely related to B/Shanghai/361/2002 and the more recent egg-grown reference strain B/Egypt/144/2005 (Table 7). The HAs and NAs of most recent isolates were similar to those of viruses circulating during 2004. Their HAs were characterized by the change V252M, many with an additional change P108A, relative to B/Shanghai/361/2002 (red in Figure 5, Table 9). As previously, their NA sequences differed in some 10 amino acids from the NA of B/Shanghai/361/2002 (green in Figure 7, Table 9). A few fell within the phylogenetic group represented by B/Jiangsu/10/2003, their HAs characterized by the amino acids changes K129N, I180V or T182K (+G229S), D232A and G256R, relative to B/Shanghai/361/2002 (blue in Figure 5, Table 9). Their NA sequences were similar to that of B/Shanghai/361/2002 (blue in Figure 7, Table 9).

Many of the B/Victoria-lineage (HA) viruses were antigenically closely-related to B/Shangdong/7/97; however many gave reduced HI titres, especially with ferret antisera to B/Tehran/80/2002 and B/Brisbane/32/2002 and, in recent HI tests, were shown to be antigenically more closely related to the MDCK-grown reference strain B/Hong Kong/45/2005 (Tables 7 and 8). In a few recent HI tests, many were also shown to be closely related to B/Malaysia/2506/2005 (Table 8).

Most of the HA sequences fell into one of two principal phylogenetic groups (Figure 6):

1. One characterized by the amino acid changes R48E, K80R and K129N relative to B/Shandong/7/97, and represented by B/Hong Kong/45/2005 and B/Malaysia/2506/2004 (red in Figure 6, Table 9).
2. the other characterized by the changes, K80E plus V190I (represented by B/Illinois/13/2005) or A217S (represented by B/Hawaii/33/2005) (blue in Figure 6, Table 9) . Corresponding differences were noted in the limited number of NA sequences available (red in Figure 7, Table 9), indicating increased variation among these Victoria HA/Yamagata NA “reassortant” viruses. We did not observe any obvious differences in HI tests between viruses belonging to the different phylogenetic groups.

Acknowledgements

In addition to data received from the other 3 Collaborating Centres, sequence data for viruses isolated during this period were received from Influenza Centres in Austria, Denmark, France, Germany, Hong Kong, Israel, Italy, Norway, Portugual, South Africa and Sweden (indicated in Figures by suffixes).

Table 1. Influenza isolates, February 2005 - present

Month of isolation	H1N1	H3N2	B	
			Yamagata-lineage	Victoria-lineage
February				
Albania				1
Algeria		1	1	
Austria	2	4		
Belgium	2	4	1	
Bulgaria	5			
Croatia	2			2
Czech Republic	5	6		
Denmark	11	36	4	
France	7	15	7	2
Germany	5	19	2	16
Greece	4			
Hong Kong		13		8
Iran	6	5	3	2
Israel		3		
Iceland		7		
Italy	16	47	2	13
Morocco			1	
Poland		10		8
Portugal	2	33	14	1
Reunion		1		
Romania		2	1	2
Russia		9	2	1
Saudi Arabia	5	13	35	1
Serbia and Montenegro				2
Slovenia	3		1	
Spain		5	4	
Sweden		5	2	
Switzerland		3		
Turkey	2			
March				
Albania				1
Algeria	1			
Austria		1		
Belgium			3	
Bulgaria	1			
Czech Republic			7	
Denmark	5	14	11	
Finland	1	2	1	
France	2	5	3	
Germany	6	5	3	8
Iceland		1	1	
Italy		1	2	3
Latvia	1	9	8	
Norway	4	6	2	
Poland	1	1	4	5
Reunion			1	
Portugal		2	4	1
Romania	12	6	3	4
Russia	8	6	6	2
Saudi Arabia	6	26	10	
Slovakia	1	2	2	
Spain			2	
Sweden		2		
Switzerland	1	1	8	

Turkey	1			
April				
Denmark			4	
France			1	
Hong Kong		1		
Iceland		2		
Latvia	2	2	1	
Norway		1		
Romania				1
Russia		1	1	
Saudi Arabia		5		
Slovakia		2		
Spain			1	
Sweden	1	3	3	
Turkey				2
May				
Hong Kong		15		
Latvia		1	1	1
Reunion				3
Russia	3	1		
South Africa	2	1		2
Sweden		4	1	
June				
Hong Kong		8		
South Africa	1	3	1	2
Sweden		1		
July				
France		4		
July				
France		4		
Total = 793	137	386	176	94
	(17%)	(49%)	(22%)	(12%)

Table 2. Antigenic analyses of influenza A H1N1 viruses

Viruses	Isolation Date		Haemagglutination inhibition titre Post-infecton ferret sera A/Beij 262/96	A/NC 20/99	A/Eg 96/02	A/HK 2637/04	A/Neth 128/04	A/Thess 24/05
A/Beijing/262/96		Ex	**1280**	320	320	1280	640	640
A/New Caledonia/20/99		Ex	160	**640**	640	1280	640	1280
A/Egypt/96/2002 (H1N2)		Ex	160	640	**640**	640	640	640
A/Hong Kong/2637/2004		MDCK2 \2	40	640	320	**640**	320	320
A/Netherlands/128/2004		MDCK1 \2	160	640	640	1280	**2560**	2560
A/Thessaloniki/24/2005		E2 \1	160	640	640	1280	2560	**2560**
A/Iran/239/2005	unknown	MDCKx \1	160	320	640	1280	2560	2560
A/Zagreb/4467/2005	unknown	Ex	160	320	320	1280	2560	1280
A/Poland/WAW/7/2005	31.1.05	MDCKx \1	80	320	320	640	1280	1280
A/Paris/1690/2005	Feb-05	MDCK1 \1	80	320	320	640	1280	1280
A/Parma/132/2005	Feb-05	MDCK1 \2	320	320	640	2560	2560	2560
A/Prague/13/2005	2.2.05	E3	160	640	320	640	1280	—
A/Lisbon/55/2005	3.2.05	MDCK2 \1	160	320	640	1280	2560	2560
A/Austria/207244/2005	7.2.05	MDCK3 \2	320	640	640	1280	2560	—
A/Denmark/22/2005	7.2.05	MDCK2 \1	320	640	640	1280	2560	2560
A/Slovenia/478/2005	8.2.05	MDCK1 \1	40	320	640	1280	1280	—
A/Montpellier/937/2005	25.2.05	MDCKx+1 \1	160	320	640	1280	2560	2560
A/Belgium/740/2005	28.2.05	MDCK2 \1	160	320	640	1280	2560	2560
A/Sofia/504/2005	Mar-05	E3 \1	320	640	640	1280	2560	2560
A/Finland/610/2005	5.3.05	MDCK1 \1	160	320	640	1280	1280	—
A/Geneva/8619/2005	10.3.05	MDCKx \1	80	320	320	640	640	640
A/Romania/618/2005	15.3.05	E1 \1	160	320	320	1280	1280	1280
A/Norway/847/2005	16.3.05	MDCK1 \1	160	320	640	1280	2560	2560
A/Turkey/248/2005	17.3.05	MDCK2 \1	160	320	640	2560	2560	2560
A/Hessen/15/2005	21.3.05	MDCKx \1	160	640	640	1280	2560	—
A/Slovakia/393/2005	30.3.05	MDCK1 \1	160	320	640	2560	2560	2560
A/Stockholm/18/2005	7.4.05	MDCK1 \1	320	1280	640	2560	5120	5120
A/Latvia/4216/2005	19.4.05	MDCK1 \1	80	320	320	1280	1280	1280
A/Johannesburg/28/2005	11.5.05	MDCKx \1	160	640	640	1280	2560	2560
A/St Petersburg/10/2005	13.5.05	C2 \1	320	1280	640	1280	2560	2560
A/Kzasnoyarsk/46e/2005	20.5.05	E3 \1	160	640	640	1280	1280	1280
A/Johannesburg/452/2005	24.6.05	MDCKx \1	320	1280	640	1280	2560	2560

—: not done

Figure 1. Phylogenetic comparison of H1 HAs

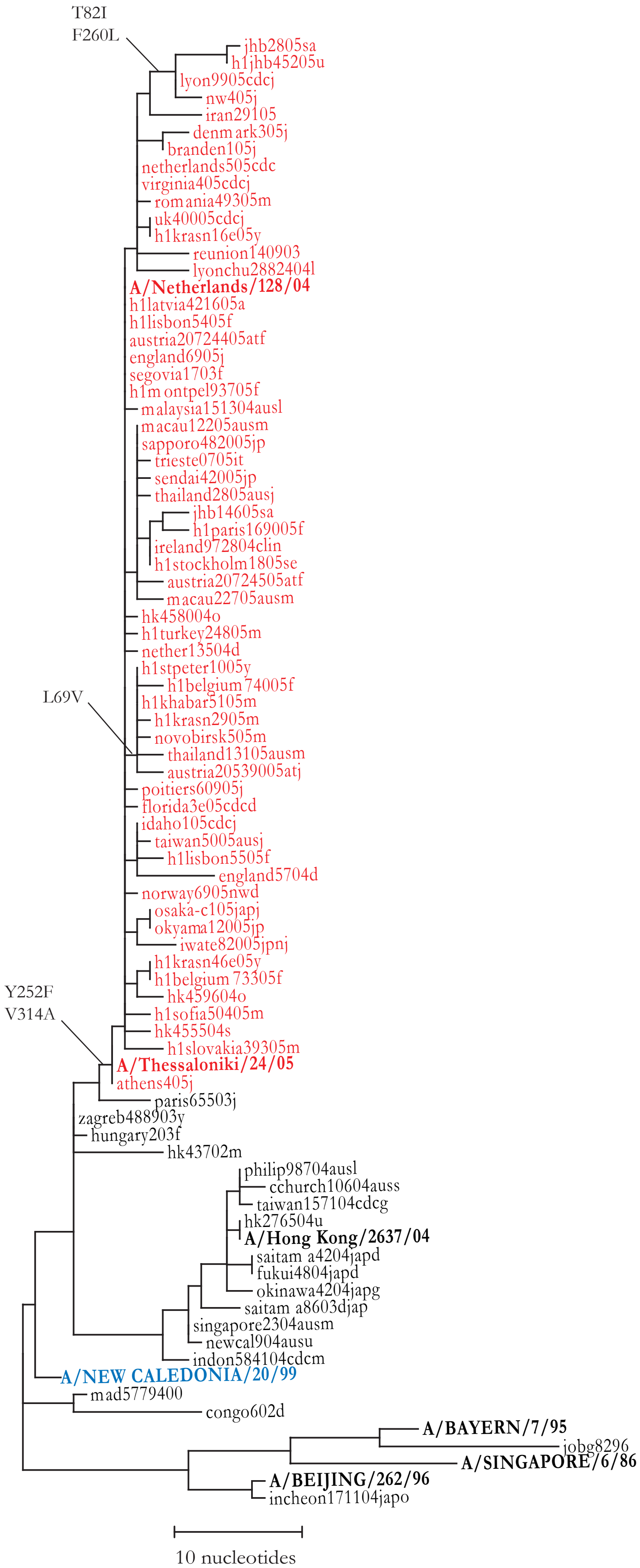

Figure 2. Phylogenetic comparison of N1s

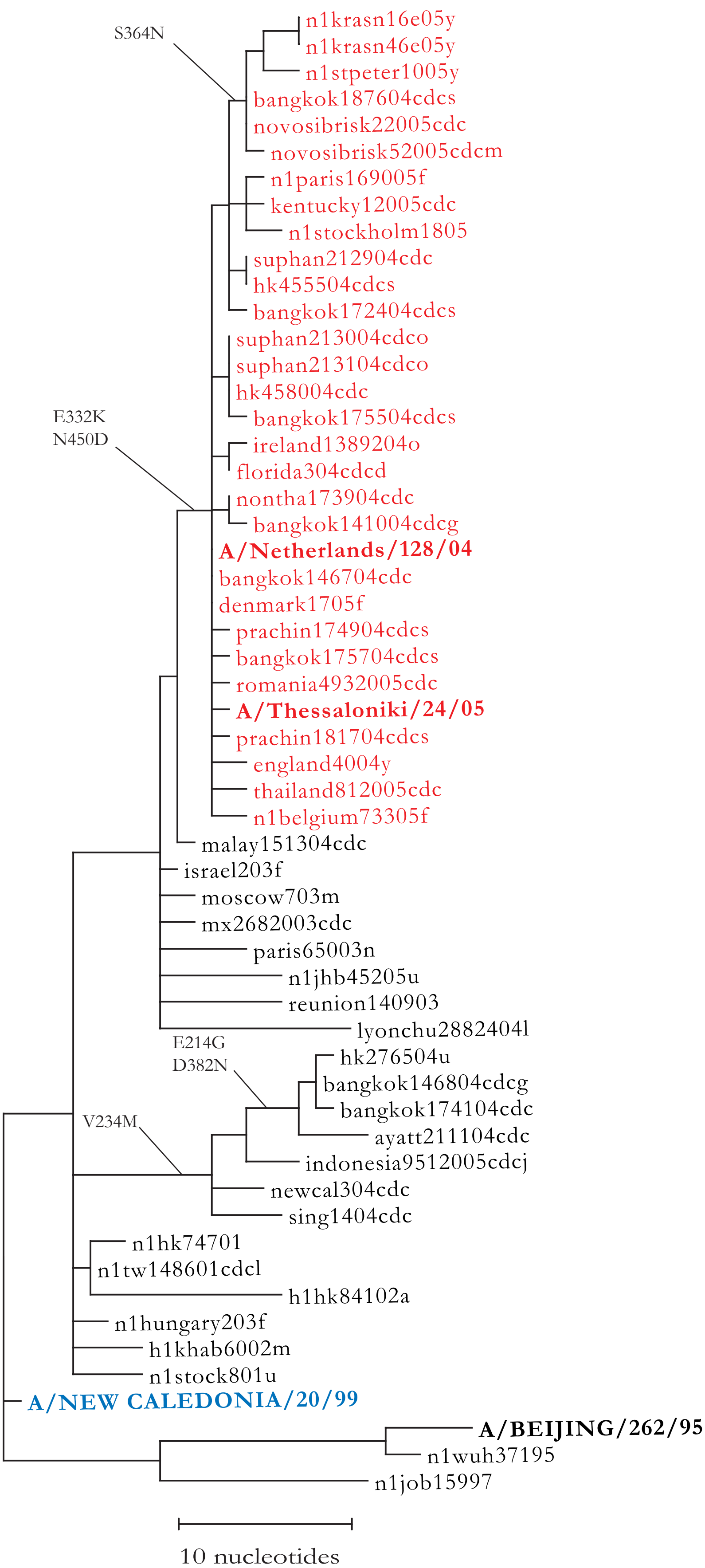

Table 3. Amino acid changes characteristic of HA and NA sequences of recent H1N1 viruses

Representative strain	Amino acid changes[1]	
	HA	NA
A/Thesaloniki/24/05	V166A W251R Y252F V314A (L69V) (T82I) (F260L)	V48I E332K N450D (S364N)

1. Relative to A/New Caledonia/20/99

Table 4. Antigenic analyses of influenza A H3N2 viruses

			Haemagglutination inhibition titre[1]							
			Post infection ferret sera							
Viruses	**Isolation Date**		**A/Pan 2007/99 F2/01**	**A/Wy 3/03 F11/03**	**A/Well 1/04 F13/04**	**A/Shan 1219/04 F14/04**	**A/Fin 486/05 F22/04**	**A/Cal 7/04 F5/05**	**A/Sing 37/04 F3/05**	**A/NY 55/05 F6/05**
A/Panama/2007/1999		Ex	**2560**	640	160	160	160	160	80	80
A/Wyoming/3/2003		Ex	1280	**5120**	640	1280	1280	2560	640	640
A/Wellington/1/2004		Ex	320	1280	**1280**	1280	1280	2560	1280	640
A/Shantou/1219/2004		MDCKx	160	640	320	**640**	320	640	640	320
A/Finland/486/05		MDCKx	160	640	160	640	**640**	640	320	640
A/California/7/2004		SpfCk3, E3	160	1280	320	2560	1280	**2560**	1280	2560
A/Singapore/37/2004		E4	160	320	320	2560	1280	2560	**2560**	2560
A/New York/55/05		Ex	160	1280	320	1280	1280	2560	1280	**2560**
A/Iran/294/2005	unknown	MDCKx \1	40	320	__	__	320	640	320	__
A/Agadir/422/2005	12.1.05	MDCKx \1	160	1280	320	1280	1280	1280	640	__
A/Slovenia/146/2005	18.1.05	MDCK2 \2	40	320	80	320	320	640	320	640
A/Athens/605	21.1.05	MDCK2 \1	160	640	320	1280	1280	1280	2560	__
A/Bratislava/84/2005	26.1.05	MDCK1 \1	320	320	160	640	640	1280	1280	__
A/Israel/64/2005	31.1.05	MDCK1 \1	80	640	160	1280	640	2560	640	2560
A/Paris/1734/2005	Feb-05	MDCK1 \1	160	640	640	1280	640	1280	__	1280
A/Parma/84/2005	Feb-05	MDCK1 \1	80	320	160	320	640	640	320	640
A/Rome/2/2005	Feb-05	MDCK1 \1	40	320	160	320	__	320	160	320
A/Trieste/31/2005	1.2.05	MV1LU \2	80	640	160	640	640	1280	320	640
A/Hong Kong/91/2005	3.2.05	MDCK2 \1	160	640	320	1280	1280	2560	__	1280
A/Austria/207265/2005	7.2.05	MDCK3 \1	80	1280	160	1280	1280	5120	640	2560
A/Poland/WAW/17/2005	7.2.05	MDCKx \2	40	320	80	__	320	640	320	__
A/Salamanca/26/2005	8.2.05	MDCK1 \1	160	2560	320	2560	__	2560	2560	__
A/San Sebastian/RR1823/2005	9.2.05	MDCK1 \2	80	640	160	640	1280	1280	__	640
A/Prague/34/05	13.2.05	MDCK5 \2	80	320	80	640	__	640	640	__
A/Lisbon/87/2005	19.2.05	MDCK1 \2	80	320	80	640	__	640	160	640
A/Denmark/55/2005	21.2.05	MDCK2 \1	160	1280	640	1280	1280	2560	1280	2560
A/Algeria/G04/2005	23.2.05	MDCK2 \1	80	1280	320	1280	__	1280	640	2560
A/Belgium/739/2005	24.2.05	MDCK1 \2	80	320	80	640	__	1280	320	1280
A/Saudi Arabia/384/2005	24.2.05	MDCK1 \1	80	640	160	640	1280	1280	__	1280
A/Finland/603/2005	29.2.05	MDCK2 \1	160	640	320	1280	1280	2560	__	2560
A/Geneva/8379/2005	4.3.05	MDCK2 \1	80	320	320	__	640	1280	640	640
A/Norway/855/2005	14.3.05	MDCK1 \1	160	640	320	__	640	2560	640	640
A/Latvia/3010/2005	17.3.05	MDCK2 \1	80	640	160	640	__	1280	640	640
A/Lyon/CHU/110508/2005	17.3.05	MDCK2 \1	80	640	160	1280	__	1280	640	__
A/Berlin/94/05	21.3.05	MDCKx \1	160	320	__	1280	1280	1280	640	1280
A/Romania/810/2005	30.3.05	MDCK2 \1	80	1280	640	640	640	1280	640	1280
A/Iceland/111/2005	16.4.05	MDCK2 \1	80	320	160	640	__	640	640	320
A/Slovakia/44/2005	19.4.05	MDCKx \1	40	320	160	640	__	640	320	__
A/Stockholm/19/2005	19.4.05	MDCK1 \1	80	1280	320	1280	640	2560	__	1280
A/St Petersburg/8/2005	22.4.05	C3 \1	40	160	80	160	320	640	__	640
A/Hong Kong/2782/2005	20.5.05	MDCK2 \1	<	80	80	80	__	160	160	160
A/Johannesburg/122/2005	30.5.05	MDCKx \2	40	80	80	80	80	160	__	160
A/Stockholm/20/2005	8.6.05	MDCK1 \1	40	80	80	160	160	160	__	160
A/Johannesburg/391/2005	20.6.05	MDCKx \3	40	160	80	160	160	320	__	320
A/Hong Kong/4443/2005`	24.6.05	MDCK2 \1	<	80	80	80	80	160	__	160
A/Annecy/1140/2005	1.7.05	MDCK3 \1	<	80	80	160	__	320	__	160

1. <,<40; --, not done

Table 5. Antigenic analyses of influenza A H3N2 viruses

			Haemagglutination inhibition titre[1]									
			Post-infection ferret sera									
Viruses	**Isolation Date**		**A/Pan 2007/99 F2/01**	**A/Wy 3/03 F11/03**	**A/Shan 1219/04 F14/04**	**A/Fin 486/04 F22/04**	**A/Cal 7/04 F5/05**	**A/NY 55/04 F6/05**	**A/BW 38/05 F20/05**	**A/NH 14/05 CDC**	**A/Bris 3/05 AUS**	**A/HK 4355/05 F23/05**
A/Panama/2007/1999		Ex	**2560**	320	80	80	80	40	80	40	40	80
A/Wyoming/3/2003		Ex	640	**5120**	640	1280	2560	640	320	1280	160	640
A/Shantou/1219/2004		MDCKx	160	640	**640**	640	640	640	320	640	320	640
A/Finland/486/2004		MDCK6	40	320	640	**640**	1280	640	320	640	160	640
A/California/7/2004		SpfCk3, E3	80	1280	1280	1280	**2560**	1280	1280	1280	320	1280
A/New York/55/2004		Ex	40	320	1280	640	1280	**1280**	1280	1280	320	640
A/Baden-Wurttemburg/38/2005		MDCK5	40	320	640	640	640	640	**640**	640	160	320
A/New Hampshire/14/2005		SpfCK3E2 \1	80	640	1280	1280	2560	1280	640	**1280**	640	320
A/Brisbane/3/2005		E4 \1	80	320	2560	1280	2560	1280	640	1280	**1280**	640
A/Hong Kong/4355/2005		MDCK2 \3	40	80	80	160	320	160	40	160	160	**640**
A/Gunma/16/2005	unknown	MDCK3 \1	160	320	320	160	160	160	320	160	80	—
A/Victoria/503/2005	unknown	E4 \1	40	640	1280	1280	1280	1280	640	1280	640	—
A/Brisbane/23/2005	unknown	E3 \1	<	320	640	640	640	640	320	640	640	—
A/Christchurch/64/2005	unknown	MDCKx/1 \2	80	160	320	160	160	80	160	160	80	160
A/Macau/557/2005	unknown	MDCKx/5 \2	<	40	40	80	80	80	<	40	40	<
A/Yaroslavl/13/2005	1.2.05	MDCK1 \3	40	80	320	320	320	320	80	160	80	320
A/Moscow/9/2005	21.2.05	MDCKx \3	<	80	160	160	320	160	40	160	80	320
A/Saudi Arabia/569/2005	9.3.05	MDCK1 \1	80	640	1280	2560	2560	1280	1280	2560	640	640
A/Saudi Arabia/50/2005	13.3.05	MDCK1 \1	40	80	320	320	320	160	80	160	320	160
A/Saudi Arabia/521/2005	23.3.05	MDCK1 \1	<	80	160	160	320	320	80	320	160	160
A/Saudi Arabia/340/2005	2.4.05	MDCK2 \1	40	160	640	320	640	640	160	320	160	160
A/Saudi Arabia/441/2005	2.4.05	MDCK3 \1	<	80	320	320	640	640	80	320	160	160

1. <,<40

Figure 3. Phylogenetic comparison of H3 HAs

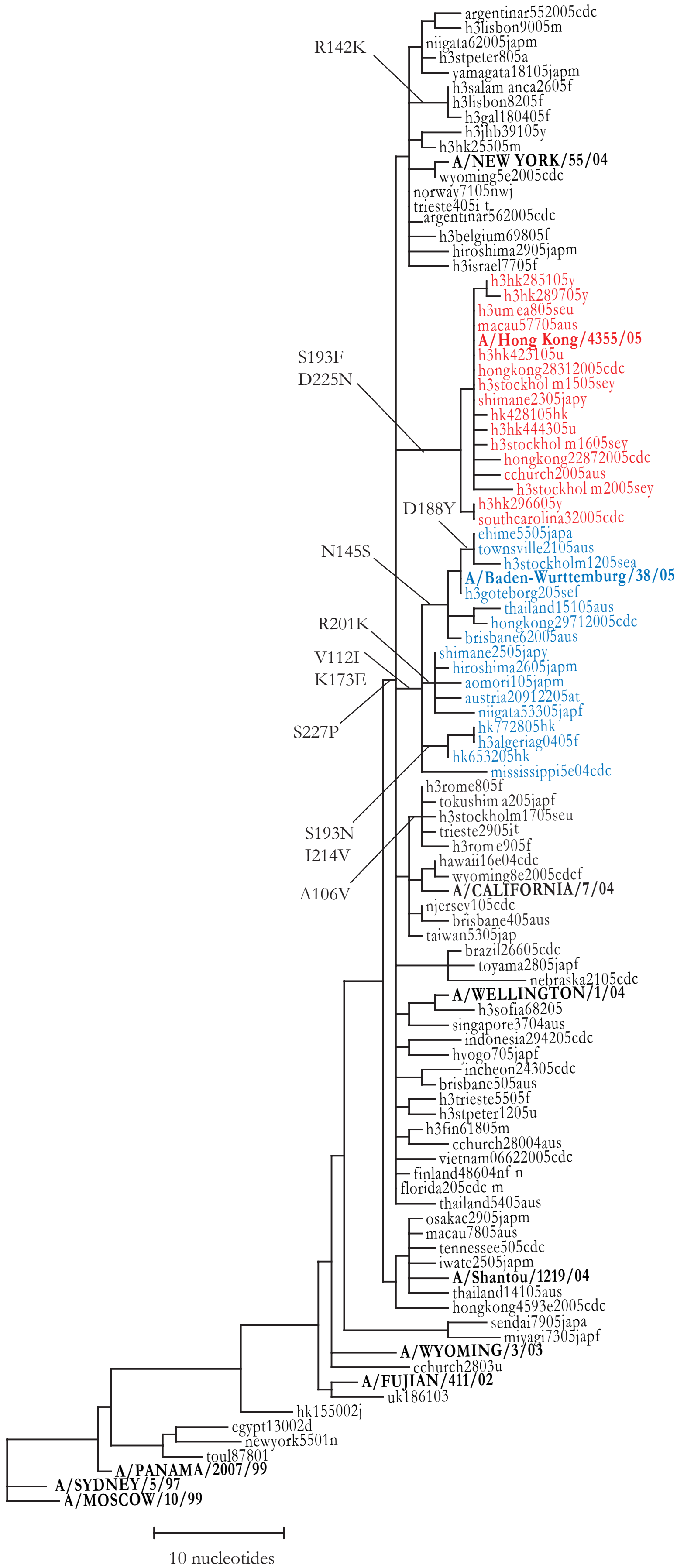

Figure 4. Phylogenetic comparison of N2s

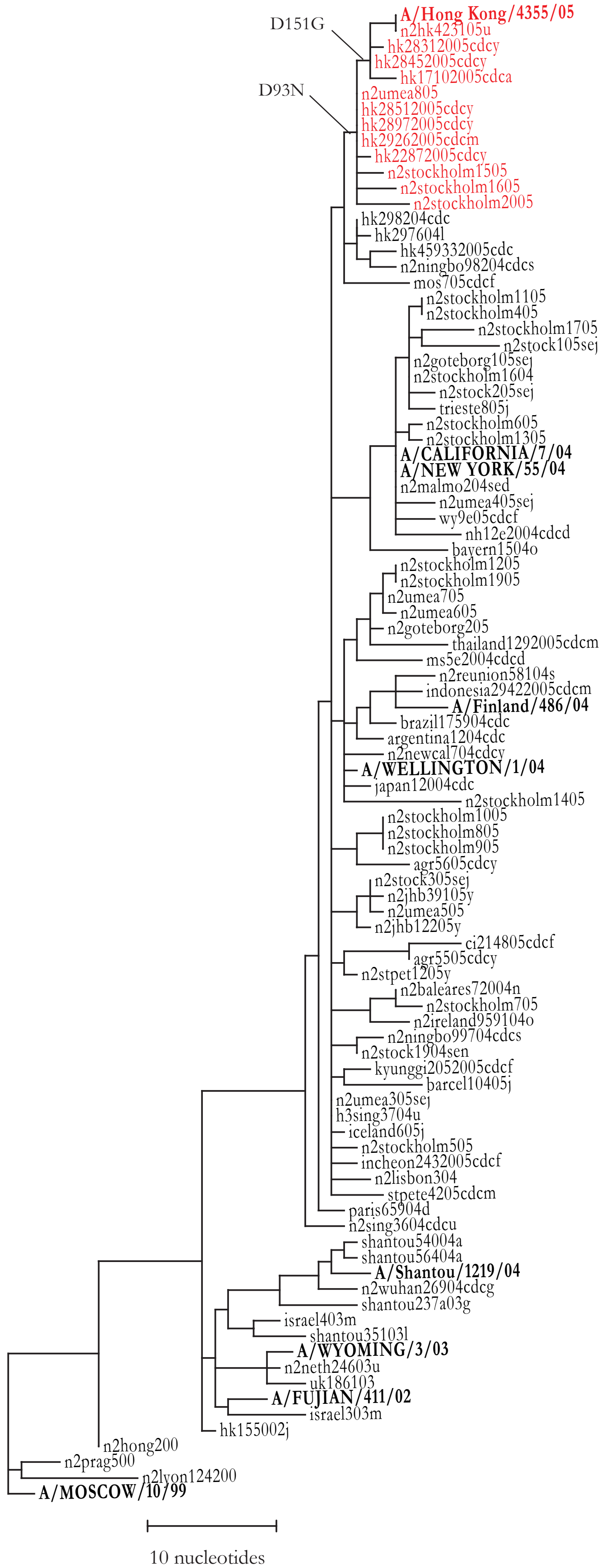

Table 6. Amino acid changes characteristic of H3N2 sequence variants

Variant group	Representative strain	Amino Acid changes[1]	
		HA	NA
1	A/Shantou/1219/04	P227S K326T	D93N I215V E221K K431N E432Q
2	A/BW/38/05	V112I K173E (N145S, D188Y) (R201K) (S193N, I214V)	(E221K)
3	A/Hong Kong/4355/05	S193F D225N	D93N (D151G)

1. Relative to A/California/7/04.

Table 7. Antigenic analyses of influenza B viruses

Viruses	Isolation date		Haemagglutination inhibition titre[1]								
				Post-infection ferret sera							
			B/Shan[2] 7/97	B/Shan 7/97	B/Tehr 80/02	B/Bris 32/02	B/HK 45/05	B/Sich 379/99	B/Shai 361/02	B/Jiang 10/03	B/Egypt 144/05
B/Shandong/7/97		**Ex**	**640**	**320**	160	160	160	<	<	<	<
B/Tehran/80/2002		**Ex**	320	160	**320**	160	160	<	<	<	<
B/Brisbane/32/2002		**Ex**	640	160	80	**160**	160	<	<	<	<
B/HK/45/2005		**MDCKx**	320	80	<	40	**320**	<	<	<	<
B/Sichuan/379/99		**Ex**	<	<	<	<	<	**160**	160	<	160
B/Shanghai/361/2002		**Ex**	<	<	<	<	<	160	**320**	40	320
B/Jiangsu/10/2003		**Ex**	<	<	<	<	<	80	80	**320**	80
B/Egypt/144/2005		**Ex**	<	<	<	<	<	160	320	40	**320**
B/Thuringen/1/2005	3.2.05	MDCKx \1	<	<	<	<	__	80	160	160	320
B/Madrid/G2627/2005	7.2.05	MDCK1 \1	<	<	<	<	__	80	160	160	320
B/Lisbon/23/2005	8.2.05	MDCK1 \2	<	<	<	<	<	160	320	160	160
B/Slovenia/645/2005	17.2.05	MDCK1 \1	<	<	<	<	__	80	160	320	320
B/Finland/614/2005	Mar-05	MDCK1 \1	<	<	<	<	__	80	160	80	160
B/Trieste/23/2005	4.3.05	MDCKx \1	<	<	<	<	__	40	80	80	160
B/Poland/Lodz/6/2005	9.3.05	MDCKx \1	<	<	<	<	__	80	160	80	160
B/Belgium/864/2005	10.3.05	MDCK2 \1	<	<	<	<	<	80	160	80	80
B/Geneva/8624/2005	10.3.05	MDCKx \1	<	<	<	<	__	160	80	160	160
B/Norway/854/2005	14.3.05	MDCK1 \1	<	<	<	<	__	80	80	160	160
B/Iceland/109/2005	17.3.05	MDCK1 \1	<	<	<	<	<	160	160	160	160
B/Vsetin/121/2005	21.3.05	MDCK3 \1	<	<	<	<	__	160	80	160	160
B/Slovakia/376/2005	23.3.05	MDCKx \1	<	<	<	<	__	80	80	80	160
B/Romania/701/2005	25.3.05	MDCK2 \1	<	<	<	<	__	80	160	320	320
B/Denmark/9/2005	31.3.05	MDCK2 \1	<	<	<	<	__	160	80	160	320
B/Lyon/1033/2005	4.4.05	MDCK2 \1	<	<	<	<	<	80	80	80	160
B/Tula/14/2005	18.4.05	C2 \1	<	<	<	<	<	160	160	160	160
B/Latvia/4901/2005	1.5.05	MDCK1 \1	<	<	<	<	__	80	80	80	160
B/Stockholm/6/2005	5.5.05	MDCK1 \1	<	<	<	<	<	320	160	80	160
B/Zagreb/3611/2005	unknown	Ex	640	320	160	160	160	<	<	<	<
B/Parma/30/2005	Feb-05	MDCK1 \1	320	160	160	160	320	<	<	<	<
B/Paris/1705/2005	Feb-05	MDCK1 \1	320	40	<	<	160	<	<	<	<
B/Albania/2/2005	10.2.05	MDCK1 \1	640	320	160	80	__	<	<	<	<
B/Khabarovsk/14/2005	28.2.05	MDCK1 \1	320	160	<	40	320	<	<	<	<
B/Poland/Olsztyn/13/2005	Mar-05	MDCKx \2	640	160	80	40	320	<	<	<	<
B/Lisbon/32/2005	2.3.05	MDCK1 \1	320	160	40	40	320	<	<	<	<
B/Bayern/39/2005	17.3.05	MDCKx \1	640	320	80	<	__	<	<	<	<
B/Romania/700/2005	25.3.05	MDCK2 \1	320	80	<	<	160	<	<	<	<
B/Turkey/389/2005	18.4.05	MDCK2 \1	320	80	<	<	__	<	<	<	<
B/Latvia/5043/2005	4.5.05	MDCK1 \1	320	80	<	<	__	<	<	<	<

[1] <, <40; --, not done

[2] hyperimmune sheep serum

Table 8. Antigenic analyses of influenza B viruses (Victoria-lineage)

Viruses	Isolation date		Haemagglutination inhibition titre[1]							
				Post-infection ferret sera						
			B/Shan[2] 7/97	B/Shan 7/97	B/Tehr 80/02	B/Bris 32/02	B/HK 45/05	B/Mal 2506/05 AUS	B/Ill 13/05 CDC	B/Shai 361/02
B/Shandong/7/97		**Ex**	**2560**	**160**	160	80	160	320	320	<
B/Tehran/80/2002		**Ex**	2560	160	**320**	80	160	640	320	<
B/Brisbane/32/2002		**Ex**	1280	160	160	**160**	160	640	320	<
B/HK/45/2005		**MDCKx**	1280	160	<	<	**320**	160	80	<
B/Malaysia/2506/2005		**Ex**	1280	160	160	160	160	**640**	320	<
B/Illinois/13/2005		**Ex**	1280	160	160	160	160	640	**320**	<
B/Shanghai/361/2002		**Ex**	<	<	<	<	<	<	<	**320**
B/Zagreb/3611/2005	unknown	Ex	2560	320	160	160	160	640	320	<
B/Zagreb/3704/2005	unknown	Ex	2560	160	80	80	80	320	320	<
B/Victoria/505/2005	unknown	E3 \1	1280	160	160	80	160	640	640	<
B/Paris/475/2005	Dec-04	MDCK1 \1	2560	160	40	40	320	320	80	<
B/Parma/30/2005	Feb-05	MDCK1 \1	2560	160	80	40	320	160	80	<
B/Paris/935/2005	Mar-05	MDCK1 \1	1280	160	80	40	160	320	160	<
B/Poland/Olsztyn/13/2005	Mar-05	MDCKx \2	2560	160	<	<	320	160	40	<
B/Khabarovsk/57/2005	17.3.05	MDCK1 \1	2560	160	80	80	160	320	640	<
B/Johannesburg/27/2005	12.5.05	MDCKx \1	2560	160	40	40	320	160	160	<
B/Reunion/1155/2005	30.5.05	MDCKx+3 \1	2560	160	40	40	320	160	80	<
B/Johannesburg/403/2005	20.6.05	MDCKx \1	2560	80	<	<	320	40	160	<
B/Reunion/1156/2005	30.5.05	MDCKx+3 \1	2560	80	<	<	320	80	160	<

[1] <, <40;

[2] hyperimmune sheep serum

Figure 5. Phylogenetic comparison of B HAs (Yamagata lineage)

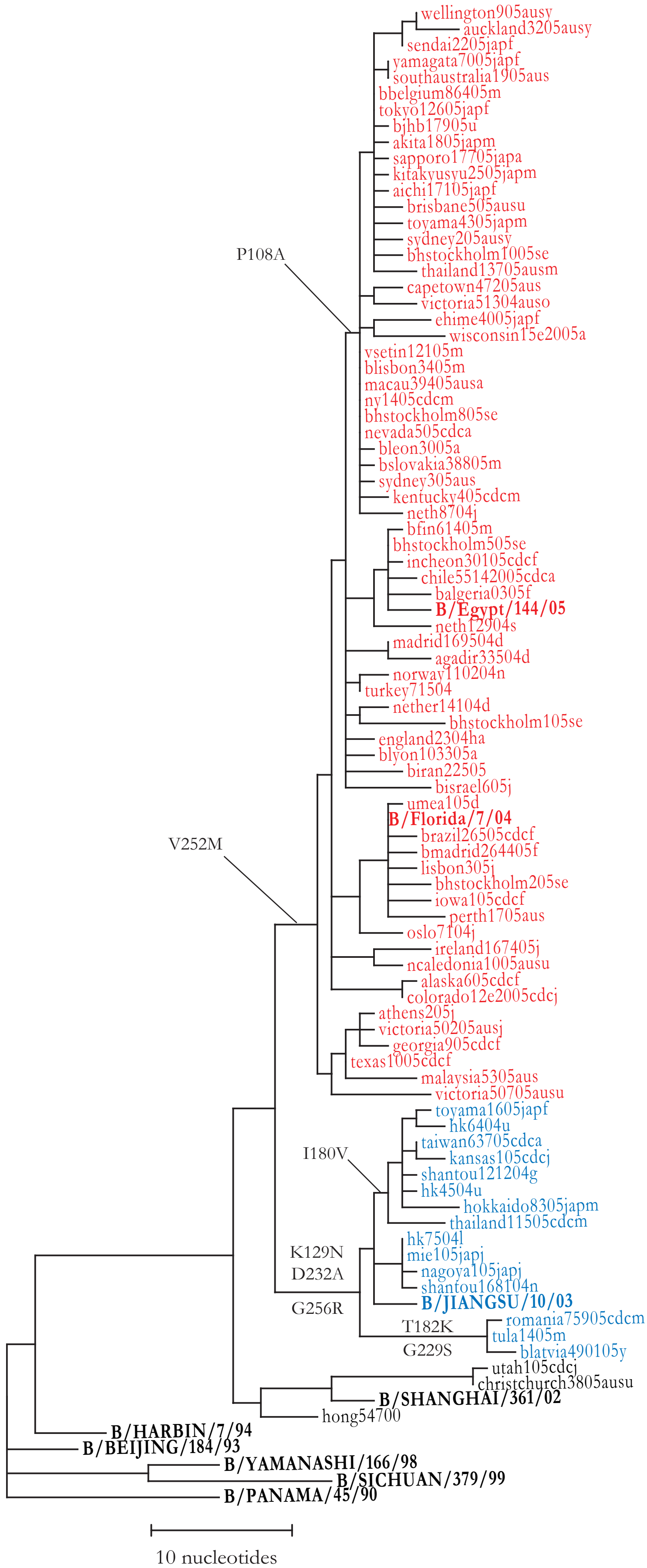

Figure 6. Phylogenetic comparison of B HAs (Victoria lineage)

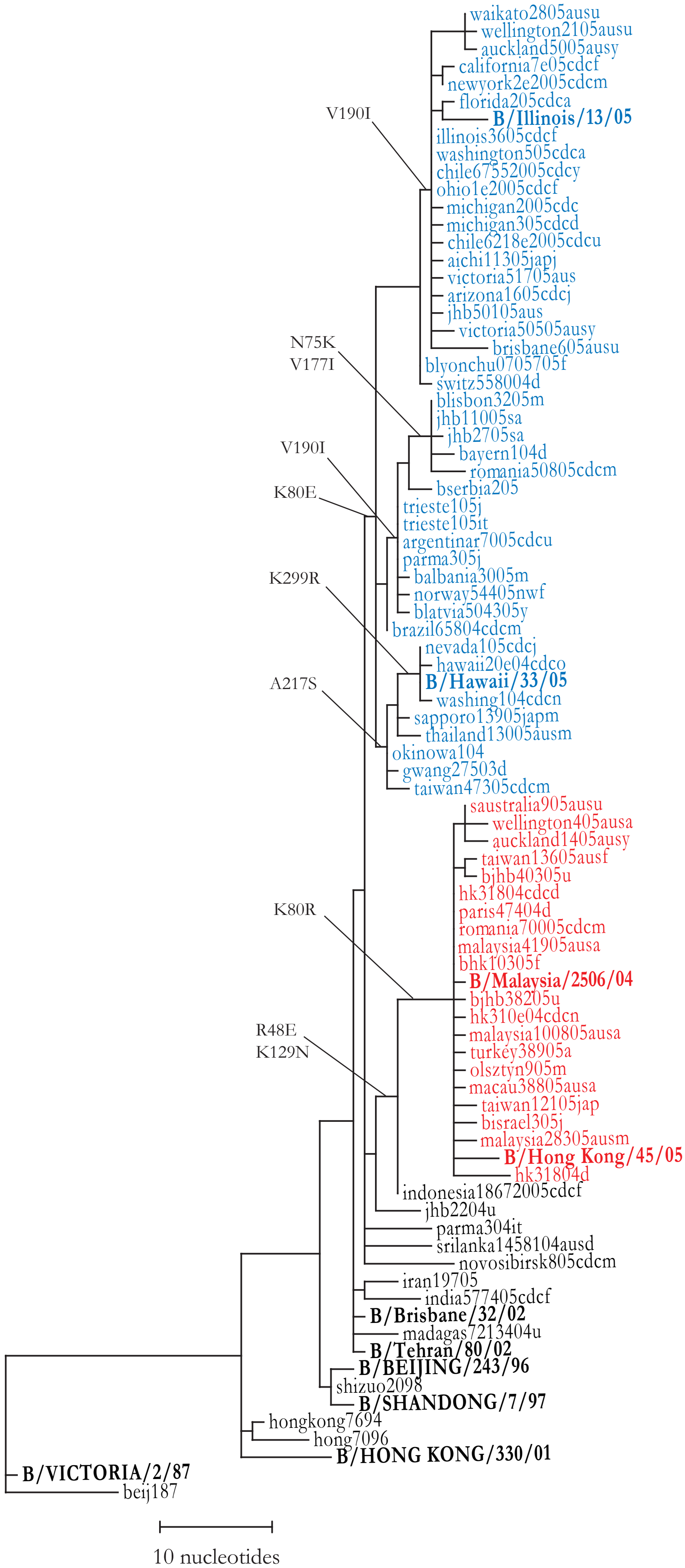

Figure 7. Phylogenetic comparison of B NAs

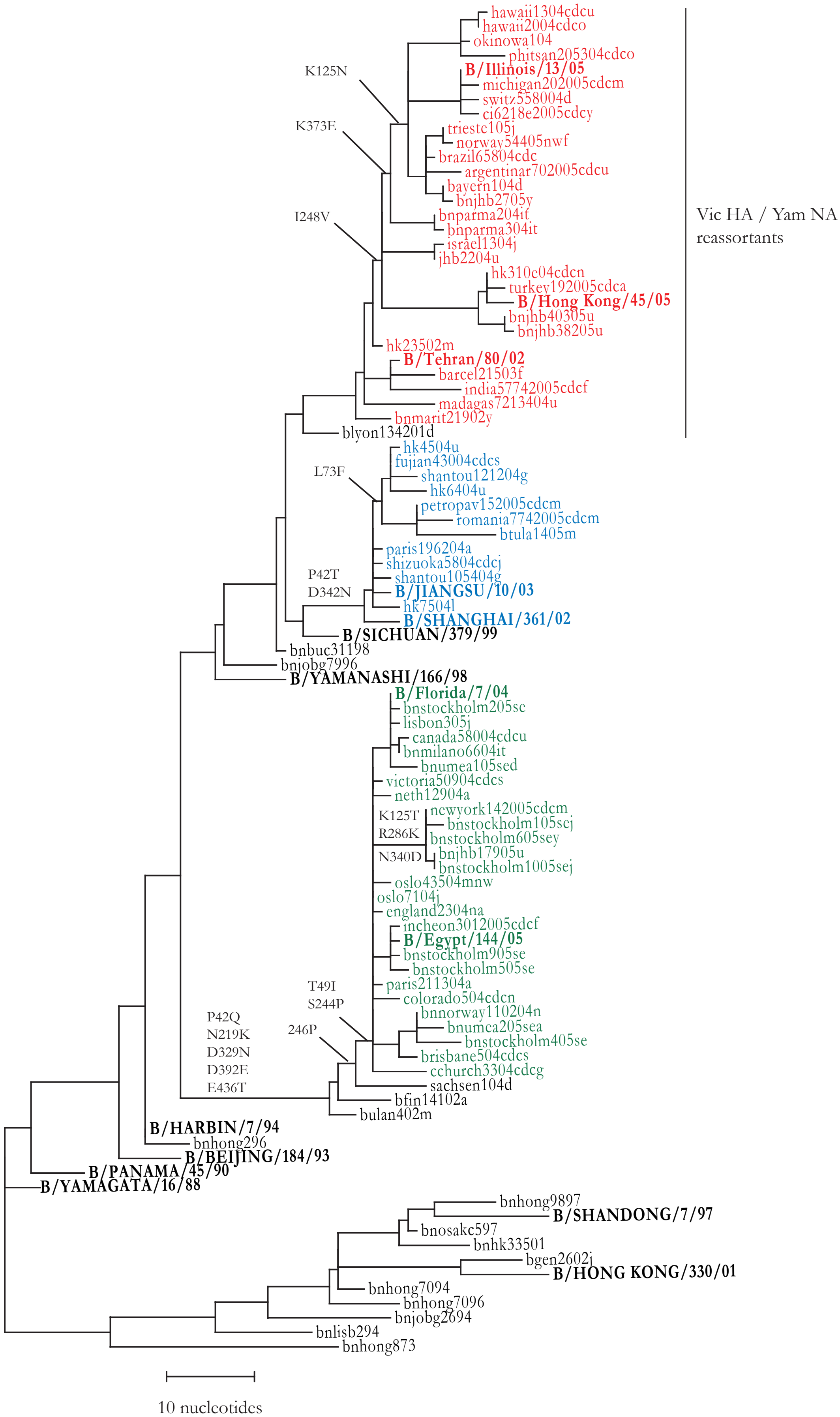

Table 9. Amino acid changes characteristic of recent B sequence variants

HA Lineage	Representative strain	Amino acid changes	
		HA	NA
Yamagata[1]	B/Jiangsu/10/03	K129N D232A G256R (I180V) (T182K)	L73F
	B/Egypt/144/05	V252M (I) (P108A)	T42Q T49I K186R N219K N235D S244P D329N N342D D392E E436T
Victoria[2]	B/Hong Kong/45/05	R48E K80R K129N	S50P P51S I248V V271I L396F D463N
	B/Hawaii/33/05	K80E A217S (K299R)	K125N I248V K373E S397I D463N
	B/Illinois/13/05	K80E V190I	K125N I248V K373E E404G

1. Relative to B/Shanghai/361/02. 2. HA relative to B/Shandong/7/97; NA relative to B/Tehran/80/02.